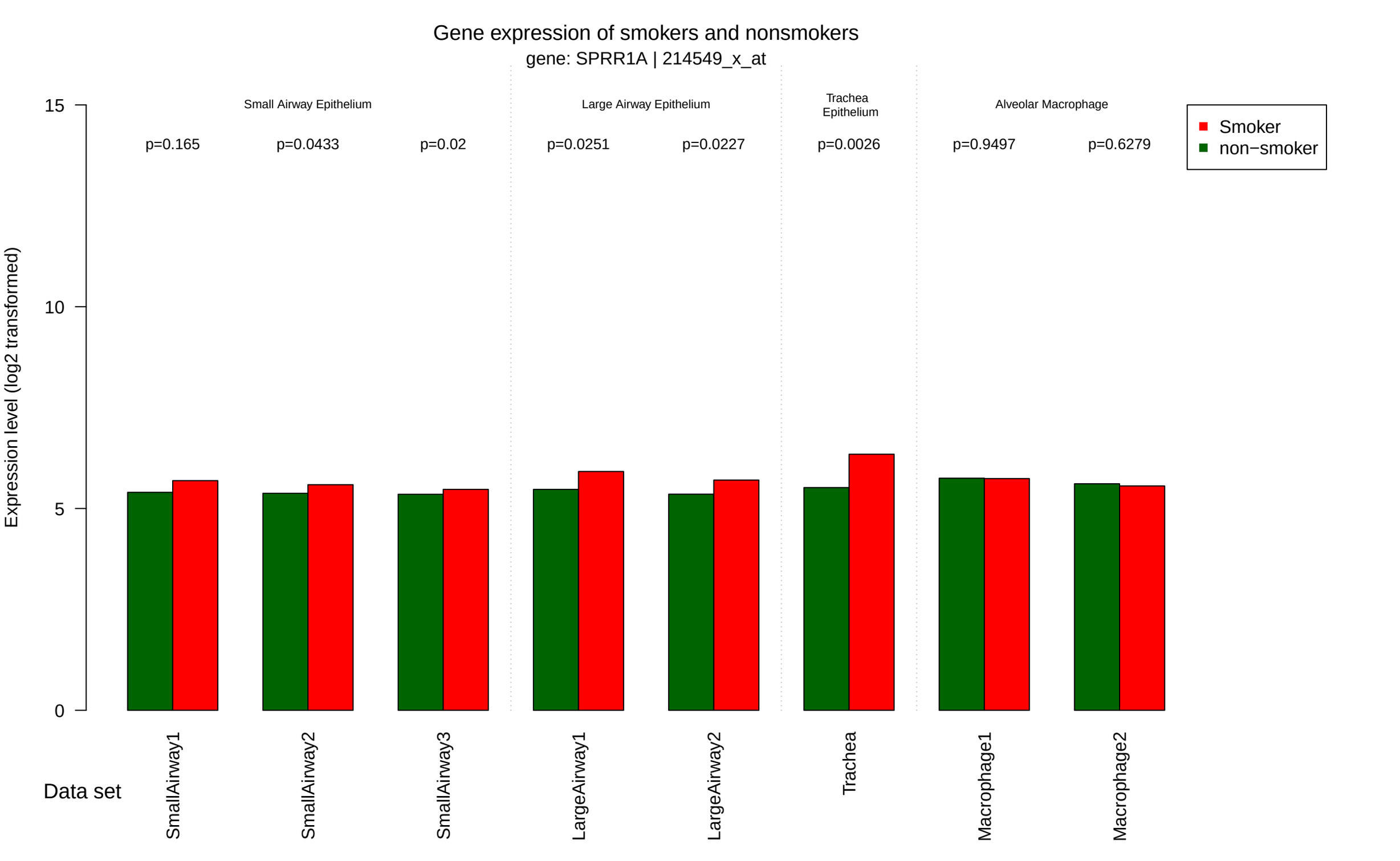
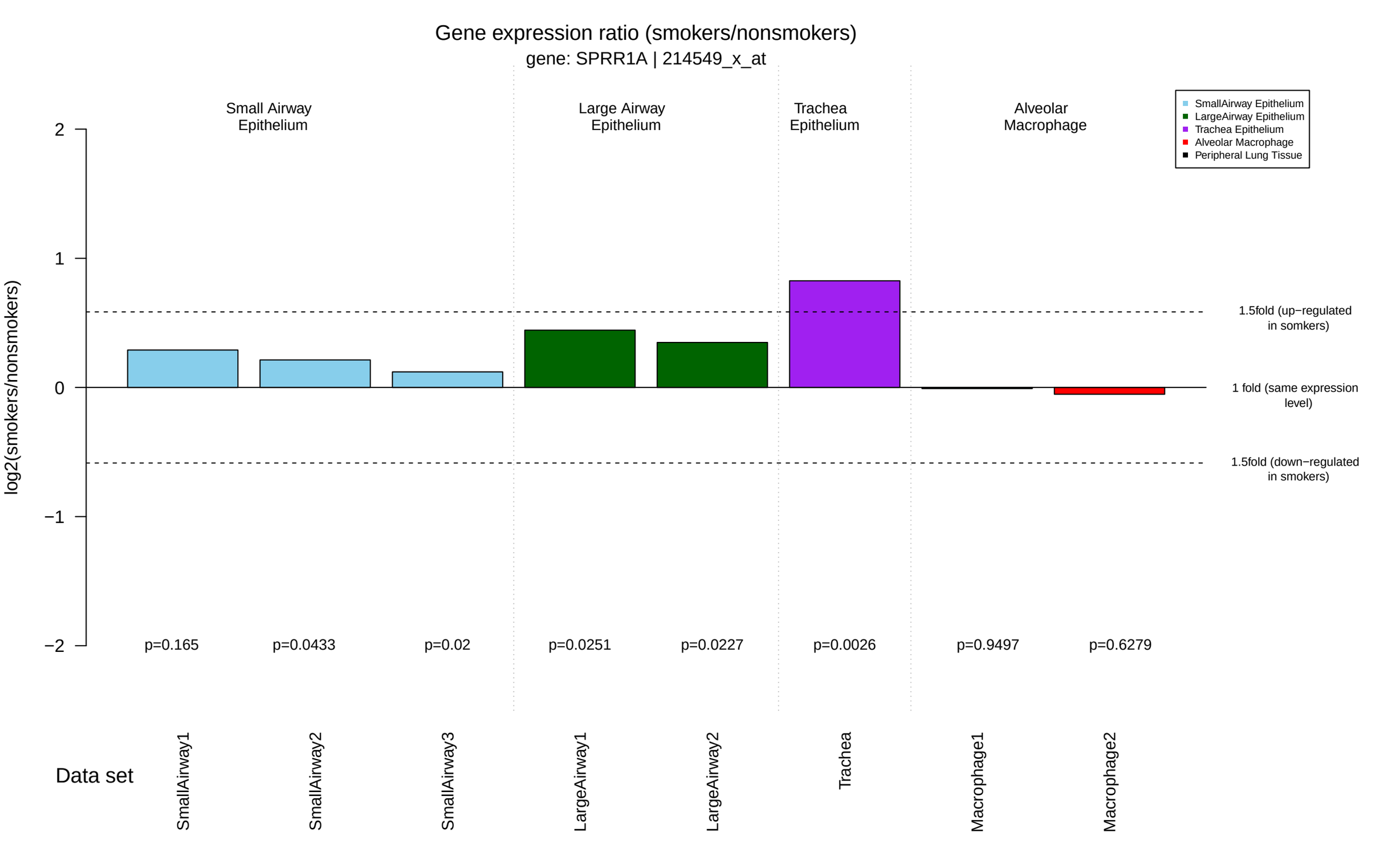
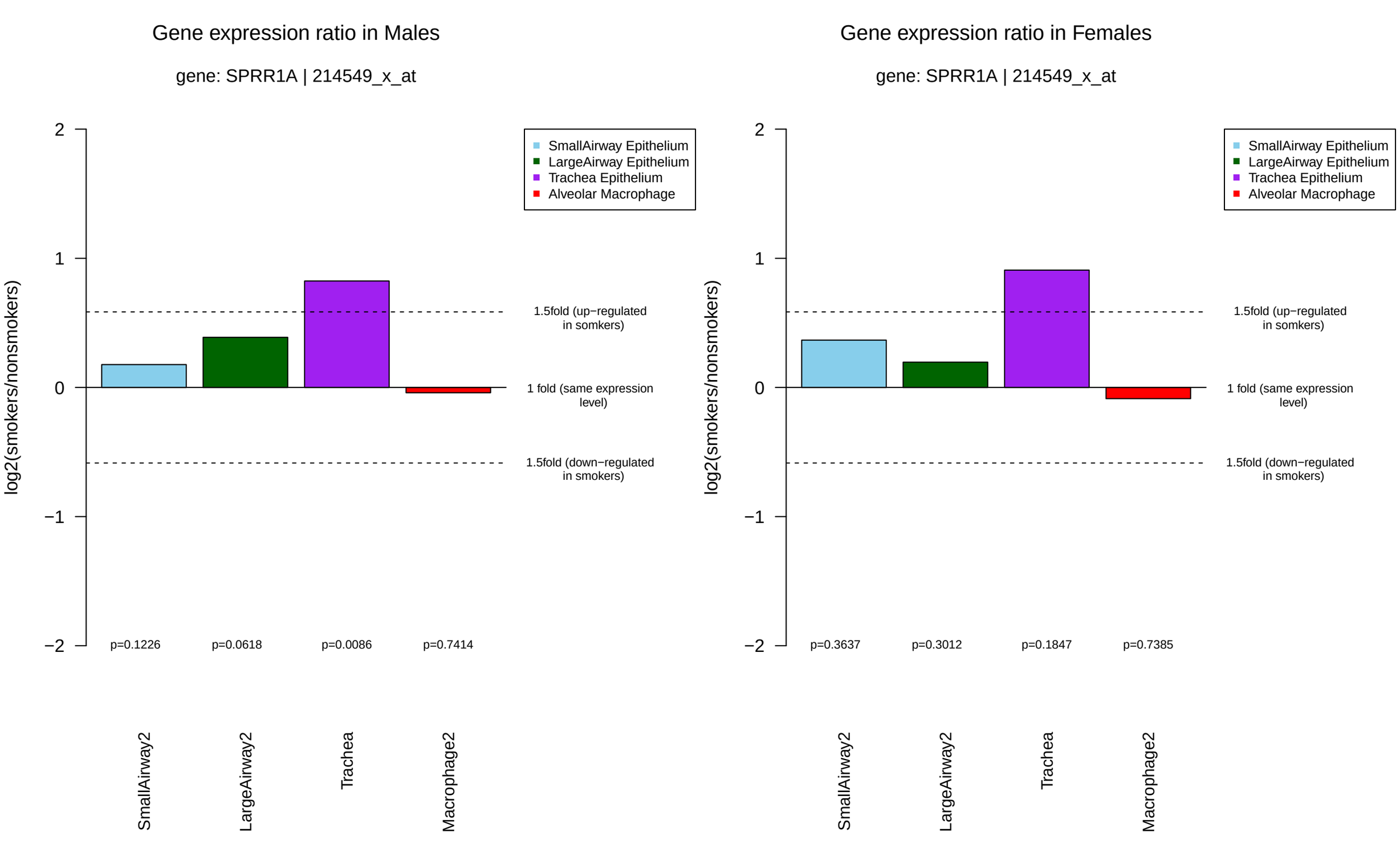
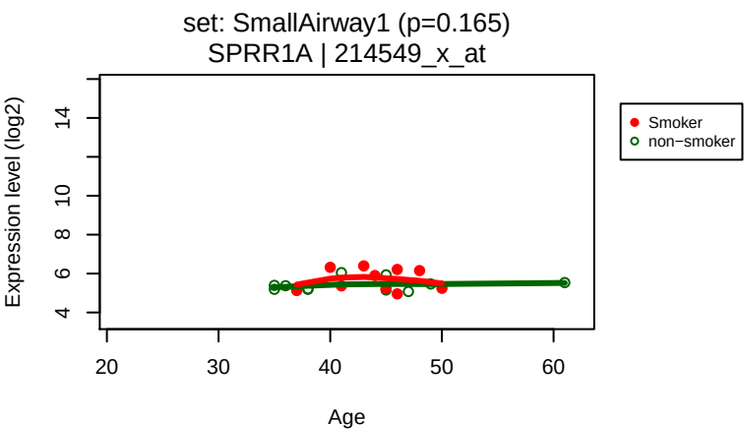
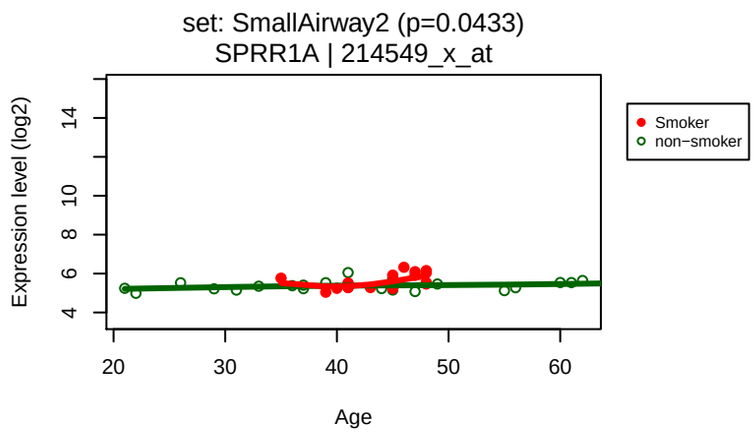
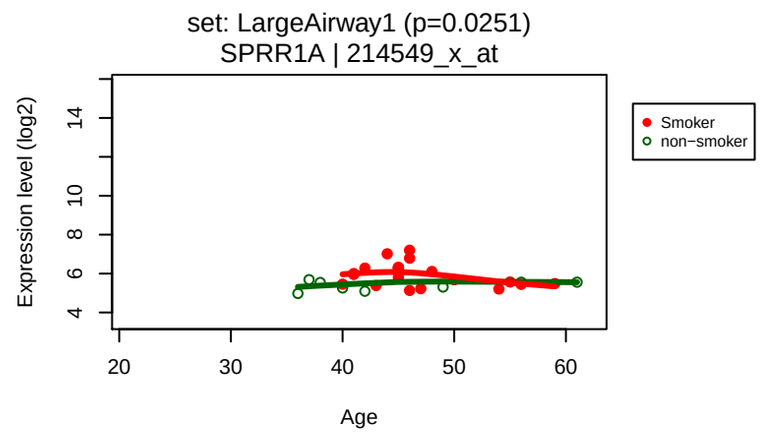

Gene expression of smokers and nonsmokers
gene: SPRR1A | 214549_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.165
p=0.0433
p=0.02
p=0.0251
p=0.0227
p=0.0026
p=0.9497
p=0.6279
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

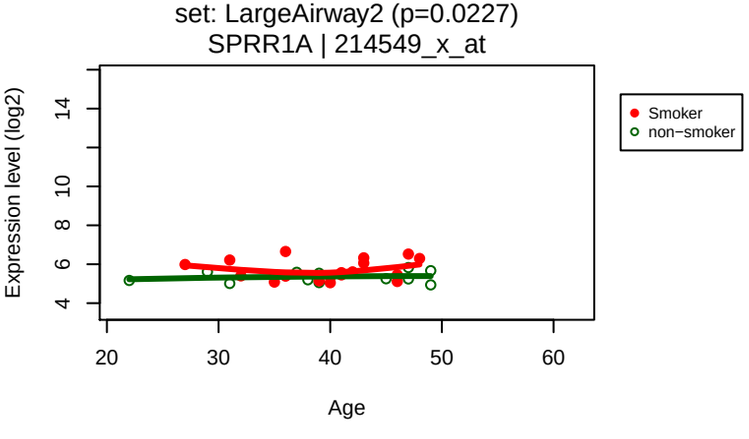

Gene expression ratio (smokers/nonsmokers)
gene: SPRR1A | 214549_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.165
p=0.0433
p=0.02
p=0.0251
p=0.0227
p=0.0026
p=0.9497
p=0.6279
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

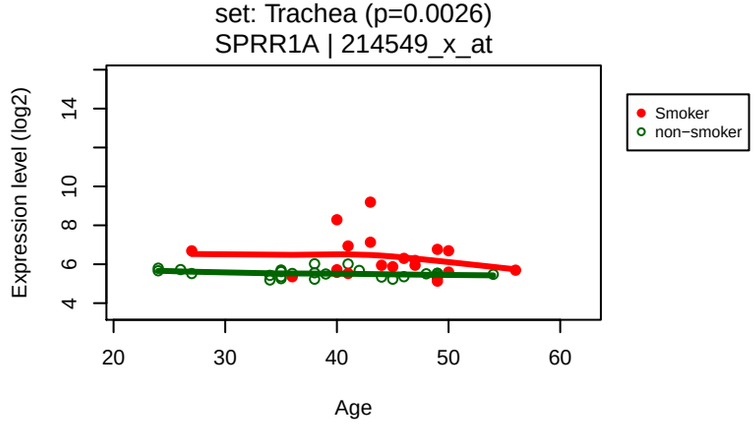

Gene expression ratio in Males
gene: SPRR1A | 214549_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1226
p=0.0618
p=0.0086
p=0.7414
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPRR1A | 214549_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3637
p=0.3012
p=0.1847
p=0.7385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

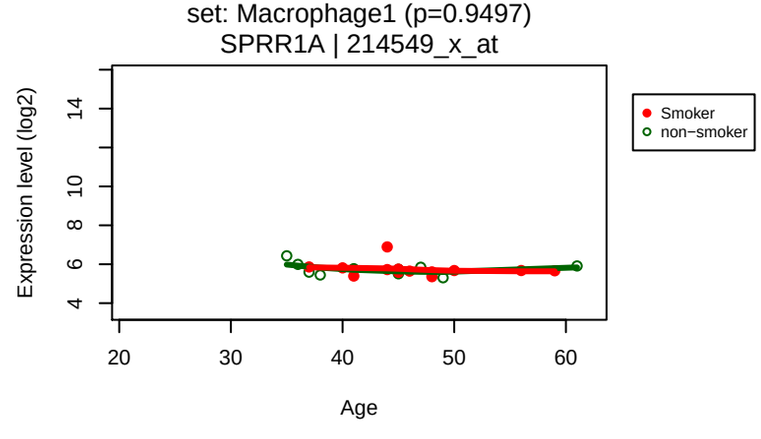

set: SmallAirway1 (p=0.165)
SPRR1A | 214549_x_at
Expression level (log2)
Age
Smoker
non−smoker

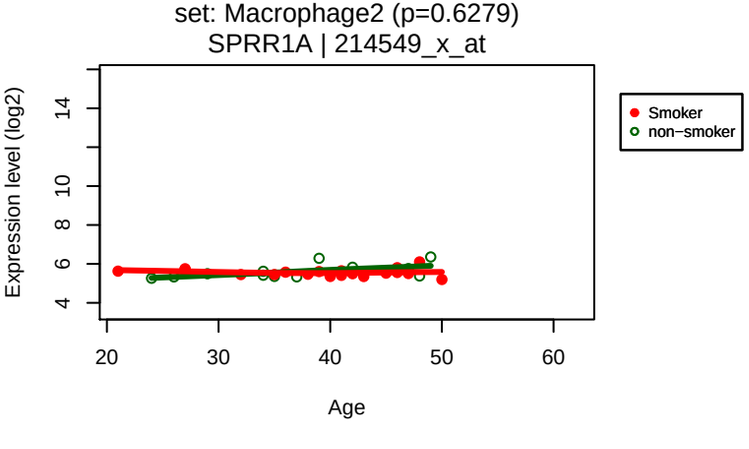

set: SmallAirway2 (p=0.0433)
SPRR1A | 214549_x_at
Expression level (log2)
Age
Smoker
non−smoker

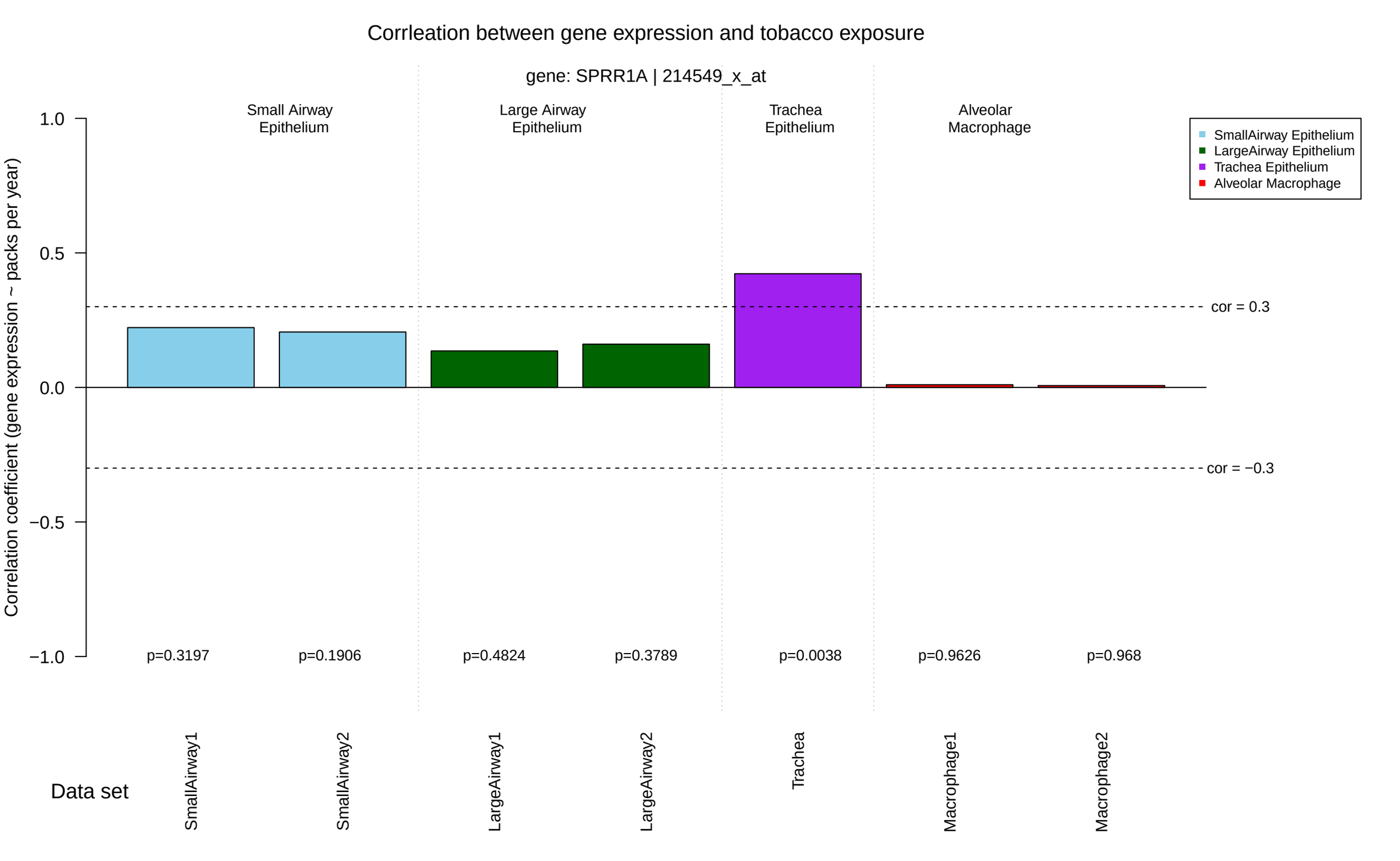

set: LargeAirway1 (p=0.0251)
SPRR1A | 214549_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0227)
SPRR1A | 214549_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0026)
SPRR1A | 214549_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9497)
SPRR1A | 214549_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6279)
SPRR1A | 214549_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPRR1A | 214549_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3197
p=0.1906
p=0.4824
p=0.3789
p=0.0038
p=0.9626
p=0.968
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2